The Institute of Chartered Accountants of Bangladesh (ICAB)

Guide Document On

INTERNATIONAL VALUATION STANDARDS

(WHAT THEY ARE AND WHAT WE SHOULD KNOW)

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Guide Document on International Valuation Standards (What They are and What We Should Know)

The Institute of Chartered Accountants of Bangladesh

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Acknowledgment: The Institute of Chartered Accountants of Bangladesh (ICAB) expresses its sincere appreciation to **Sk Md Tarikul Islam FCA (ICAB), ACA (ICAEW), CBV (Chartered Business Valuator), MBA (Bradford, UK)** for his invaluable contribution to the Guide Document on International Valuation Standards, prepared as an auxiliary resource for Strategic Business Management and Leadership.

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** Italic denotes changes*

INTRODUCTION

This document is the first publication by The Institute of Chartered Accountants of Bangladesh (ICAB) with respect to International Valuation Standards (IVS). The purpose of this publication is to raise awareness and understanding of IVS among ICAB Members, Registered Students and Corporates. This publication provides a high-level factual overview of the structure of the IVS and the content therein that is relevant to business valuation.

IVS are a global set of valuation standards promulgated by the International Valuation Standards Council (IVSC), which is an independent, not-for-profit global standard setter for the valuation profession headquartered in London (UK). The IVSC's goal is to build trust in valuation by establishing one set of globally consistent, high-quality valuation standards.

The IVSC has more than 200 member organisations from around the world, operating across **137 countries**, and is supported by numerous sponsor organisations who are leaders in the valuation field. The IVSC facilitates collaboration and cooperation among its member organisations, who are valuation professional organisations (VPOs), valuation service providers, financial services businesses, regulators, international bodies and academic institutions. The IVSC is an independent standard setter – it does not offer training or accreditation to individuals or valuation firms. The IVSC has different forms of recognition from the UN, World Bank, IMF, European Banking Authority and IASB, amongst others. Some of the world's leading regulators send observers to IVSC meetings, and IVSC also engages at the highest levels to provide input and understand regulatory issues.

IVSC acts through Board of Trustees (responsible for the governance, strategic direction and funding of IVSC), Membership & Standards Recognition Board (responsible for promotion of the valuation profession and for the review of membership process) and Standards Review Board (SRB). The Standards Review Board, along with the Technical Boards (Business Valuation Board, Financial Instruments Board, and Tangible Assets Board), is responsible for the creation and the revision of the IVS. It follows a process of public consultation approved by the Board of Trustees but otherwise has autonomy over its agenda and approval of the standards.

UNDERSTANDING IVS

IVS are developed with the aim of serving as a professional benchmark for valuers around the world with the "objective to *build confidence and public trust in valuation by producing transparent and consistent standards and securing their universal adoption and implementation for the valuation of assets across the world.*".

IVS consist of mandatory requirements that must be followed in order to state that a valuation was performed in compliance with IVS. IVS are focused on the whole valuation **engagement** (i.e. process), not just the valuation **report**.

IVS are principles-based multi-disciplinary standards; hence, they include requirements for multiple asset classes - tangible assets (such as plant, equipment and infrastructure, and real property interests), businesses, intangible assets, non-financial liabilities, inventory, development property, and financial instruments. IVS requirements apply to external as well as **internal** valuation engagements.

STRUCTURE OF IVS

This version of International Valuation Standards is published on 31 January 2024, with an effective date of 31 January 2025. The IVSC permits early adoption from the date of publication. References hereafter to IVS will be to the IVS 2025 publication. IVS include a glossary, framework, seven general standards that apply to all valuation assignments, as well as eight asset standards that include requirements related to specific types of assets.

General Standards	Assets Standards
· *IVS 100 – Valuation Framework* · IVS 101 – Scope of Work · IVS 102 – Bases of Value · IVS 103 – Valuation Approaches · IVS 104 – Data and Inputs · IVS 105 – Valuation Models · *IVS 106 – Documentation and Reporting*	· IVS 200 – Businesses and Business Interests · IVS 210 – Intangible Assets · IVS 220 – Non-Financial Liabilities · IVS 230 – Inventory · IVS 300 – Plant, Equipment and Infrastructure · IVS 400 – Real Property Interests · IVS 410 – Development Property · IVS 500 – Financial Instruments

NOTE: *Appendices, which are part of International Valuation Standards, provide additional information for certain concepts articulated. In order to provide an IVS-compliant valuation, all IVS General Standards, Asset Standards and Appendices must be followed.*

IVS GLOSSARY

The Glossary does not define basic valuation, accounting or finance terms (such as asset, liability, or cash flow) as valuers are assumed to have an understanding of such terms and the definitions of such terms may vary depending on the purpose of the valuation and the jurisdiction in which it is performed. The terms included in the Glossary are:

- **Asset/Assets-** *The right to an economic benefit.*
- **Automated Valuation Model (AVM)** - *A type of model that provides an automated calculation for a specified asset at a specified date, using an algorithm or other calculation techniques without the valuer applying professional judgement over the model, including assessing, and selecting inputs or reviewing outputs.*
- **Basis (bases) of Value** - The fundamental premises on which the reported *values* are or will be based.
- **Client** – *The person who engages the valuer for a given valuation. "Clients" may be internal (ie, valuations performed for an employer) or external (ie, when the valuer is engaged by a third-party).*
- **Cost(s) (noun)** - The consideration or expenditure required to acquire or create an *asset*.
- **Data** - *Quantitative and qualitative information available to the valuer.*
- **. Discount Rate(s)** - A rate of return used to convert a monetary sum, payable or receivable in the future, into a present value.
- **Environmental, Social and Governance (ESG)** - *The criteria that together establish the framework for assessing the impact of the sustainability and ethical practices, financial performance or operations of a company, asset or liability. ESG comprises three pillars: Environmental, Social and Governance, all of which may collectively impact performance, the wider markets and society.*
- **Equitable Value** - This is the estimated price for the transfer of an *asset* or *liability* between identified knowledgeable and willing parties that reflects the respective interests of those parties.
- **Input** – *Data, assumptions, and adjustments determined to be relevant and assessed or selected by the valuer to be used in the valuation, based upon professional judgement.*
- **Intangible Asset** – *An identifiable non-monetary asset with no physical substance.*
- **Intended Use** – *The reason(s) for which a value is developed as described in the scope of work. This is also known as intended purpose.*
- **Intended User** – *Any party identified by the client and valuer in the scope of work as users of the valuation.*
- **Investment Value** –*The value of an asset to the owner or a prospective owner given individual investment or operational objectives. This may also be known as "worth".*
- **Jurisdiction** – *The legal and regulatory environment in which a valuation is performed.*

- **Liability** – *The present obligation to transfer an economic benefit. A liability has the following two essential characteristics:*
- *it is a present obligation,*
- *the obligation requires an entity to transfer or otherwise provide economic benefits to others.*
- **Liquidation Value** – *The gross amount that would be realized when an asset or group of assets are sold from a liquidation sale, with the seller being compelled to sell as of a specific date. Liquidation value can be determined under two different premises of value (see IVS 102 Bases of Value, Appendix A60):*
- *an orderly transaction with a typical marketing period, or*
- *a forced transaction with a shortened marketing period.*
- **Market Value** - The estimated amount for which an asset or liability should exchange on the valuation date between a willing buyer and a willing seller in an arm's length transaction, after proper marketing and where the parties had each acted knowledgeably, prudently and without compulsion.
- **Must** – *Actions or procedures that are mandatory.*
- **Observable Data** – *Information that is readily available to market participants about actual events or transactions that are used in determining the value for the asset and/or liability*.
- **Price (noun)** - The monetary or other consideration asked, offered or paid for an asset or to transfer a liability. Price and value may be different.
- **Professional Judgement** - *The use of accumulated knowledge and experience, as well as critical reasoning, to make an informed decision.*
- **Professional Scepticism** - *Professional scepticism* is an attitude that includes a questioning mind and critical assessment of valuation evidence.
- **Service Organisation** – *An entity (or segment of an entity) that provides information, reports or opinions including but not limited to providing market data, credit ratings or other services to support the valuation.*
- **Should** – *The valuer is expected to comply with requirements of this type unless the valuer can demonstrate that alternative actions are sufficient.*
- **Significant** – *Any aspect of a valuation which, in the professional judgement of the valuer, greatly impacts the resultant value.*
- **Specialist** – *An individual or group of individuals possessing technical skills, experience and knowledge required to perform or assist in the valuation or the review and challenge process. A specialist can be internally employed or externally engaged.*
- **Synergistic Value** - The result of a combination of two or more assets or interests where the combined value is more than the sum of the separate values. If the synergies are only available to one specific buyer, then *synergistic* value will differ from *market value*, as the *synergistic value* will reflect particular attributes of an *asset* that are only of value to a specific purchaser. The added value above the aggregate of the respective interests is often referred to as marriage value.
- **Tangible Asset** - *A physical measurable asset such as, but not limited to, property, plant, and equipment.*
- **Valuation** - The act or process of forming a conclusion on a value as of a valuation date that is prepared in compliance with IVS.
- **Valuation Approach** - *A generic term for the use of the cost, income or market approach.*
- **Valuation Date** - *The point in time to which the valuation applies.*
- **Valuation Method** - *Within a valuation approach, a specific technique to conclude a value.*
- **Valuation Model** - *A quantitative implementation of a method in whole or in part that converts inputs into outputs used in the development of a value.*

- **Valuation Process Review** - *An analysis by the valuer to assess compliance with IVS or a component of IVS applicable as at a valuation date. This does not include an opinion on the value.*
- **Valuation Review** - *A valuation review is either a valuation process review or a value review or both.*
- **Valuation Risk** - *The possibility that the value is not appropriate for its intended use.*
- **Value (noun)** - *The valuer's quantitative conclusion on the results of a valuation process that is fully compliant with the requirements of IVS as of a valuation date.*
- **Valuer** - An individual, group of individuals or individual within an entity, regardless of whether employed (internal) or engaged (contracted/external), possessing the necessary qualifications, ability and experience to execute a *valuation* in an objective, unbiased, ethical and competent manner. *In some jurisdictions, licensing is required before one can act as a valuer.*
- **Value Review** – *An analysis by the valuer applying IVS to assess and provide an opinion on the value of another valuer's work. This does not include an opinion on the valuation process.*
- **Weight**– The amount of reliance placed on a particular indication of *value* in reaching a conclusion of value.

IVS FRAMEWORK

The IVS 100 Valuation Framework consists of general principles with regard to ethics, competency, compliance, professional scepticism, and quality control. The Framework also clarifies that when a statement is made that a valuation has been undertaken in accordance with IVS, it is implicit that it has been prepared in compliance with all relevant standards issued by the IVSC. The Framework (as well as the IVS definition of Market Value) specifically state that the standards can be applied to the valuation of both assets and liabilities and present and future claims on assets and liabilities

Valuer Principles

Ethics

The valuer must follow the ethical principles of integrity, objectivity, impartiality, confidentiality, competence, and professionalism to provide a non-biased valuation and to promote and preserve the public trust. There is a fundamental expectation that controls and procedures are in place to ensure objectivity in the valuation process.

Competency

Valuations must be prepared by an individual, group of individuals or individual within an entity, regardless of whether employed (internal) or engaged (contracted/external), possessing the necessary qualifications, ability and experience to execute a valuation in an objective, unbiased, ethical and competent manner and having the appropriate technical skills, experience and knowledge of the subject of the valuation, the market(s) in which it trades and the purpose of the valuation.

Professional Skepticism

The valuer must apply an appropriate level of professional skepticism at every stage of the valuation, including a questioning mind and critical assessment of valuation evidence.

Compliance and Departures

In some circumstances a valuer may depart from IVS due to legislative, regulatory or other authoritative requirements and still state that the "valuation was performed in accordance with IVS", but must disclose the specific requirements and the significant ways in which they differ from IVS. *Such requirements should be prioritized, explained, documented, and reported in order to remain compliant with IVS.*

In such case, a valuation can still be IVS compliant; however, the valuer is required to disclose the limitations transparently. According to IVS, departures that are not a result of legislative, regulatory or other authoritative requirements are **not permitted**, and result in valuations that are **non-compliant with IVS.**

GENERAL STANDARDS

The *seven* general standards set forth the requirements for the conduct of all valuation assignments, including establishing the terms of a valuation engagement, bases of value, valuation approaches and methods, and reporting. The general standards are:

- **IVS 100 – *Valuation Framework***
- **IVS 101 – Scope of Work**
- **IVS 102 – Bases of Value**
- **IVS 103 – Valuation Approaches**
- **IVS 104 – Data and Inputs**
- **IVS 105 – Valuation Models**
- **IVS 106 – *Documentation and Reporting***

IVS 100 – VALUATION FRAMEWORK

IVS 100 Valuation Framework establishes the foundational principles that apply to all valuations. It sets out the mandatory valuer principles, quality control requirements, guidance on using specialists, and compliance obligations that must be followed for any valuation to be IVS-compliant.

IVS 100 applies to all valuation assignments regardless of the asset type, purpose, or whether the valuer is internal or external.

Valuer Principles

Principle	*Requirement*
Ethics	*The valuer must follow integrity, objectivity, impartiality, confidentiality, competence, and professionalism.*
Competency	*The valuer must have the technical skills, knowledge and experience to complete the valuation.*
Compliance	*The valuer must disclose that IVS were used and complied with.*
Professional Scepticism	*The valuer must apply a questioning mind at every stage of the valuation.*

Valuation Process Quality Control
There must be quality controls around the valuation process to ensure valuations are performed objectively, transparently, without bias and in compliance with IVS. Controls must be documented and periodically assessed.

Use of a Specialist or Service Organisation
If the valuer lacks necessary skills or knowledge, it is acceptable to seek assistance from a specialist or service organisation. The valuer must assess their capability, document it, and understand their work to rely on it.

Compliance
To be IVS compliant, the valuation must meet all General Standards, Appendices, and applicable Asset Standards. If legal or regulatory requirements conflict with IVS, they should be prioritised but must be explained and documented. Any other deviations render the valuation non-compliant.

IVS 101 – SCOPE OF WORK

Also referred to as the "terms of the engagement", the scope of work describes the fundamental terms of a valuation, such as the asset(s) being valued, the purpose of the valuation and the responsibilities of the parties involved.

IVS 101 applies to a wide spectrum of valuation assignments, including "in-house" valuations (valuations performed by valuers for their own employers), "third-party" valuations, and valuation reviews. *A scope of work is required for all valuations and valuation reviews whether the values are for internal or external use.*

The general requirement for the scope of work is that the work undertaken must be appropriate for the intended purpose. *If, in the valuer's professional judgement, the scope of work is overly restrictive, then this may not result in an IVS-compliant valuation. The client and the valuer must agree on the scope of work and that the valuation or valuation review scope is appropriate for the intended use.*

The general requirement for the scope of work is that the work undertaken must be appropriate for the intended purpose. A valuer must also ensure that the intended recipient of a valuation understands what is being provided, and must communicate the scope of work to the client **in writing before completing the assignment**, including:

Scope of Work Item	*IVS Requirement*
· *Identity of the valuer*	*The valuer must disclose any potential conflict of interest or bias.*
· *Identity of the client(s)*	*The person, persons, or entity who appoints the valuer.*
· *Identity of the intended users*	*Any party identified by the client as a user of the valuation.*
· *Asset(s)* ***and/or liability(ies)*** *being valued*	*The subject asset(s) and/or liability(ies) must be clearly identified.*
· *Valuation currency*	*The currency for the valuation and final report must be established.*
· ***Intended use*** *of the valuation*	*The reason for which a valuation is developed.*
· *Basis/bases of value used*	*Must be appropriate for the intended use; source must be cited.*
· *Valuation date*	*The point in time to which the valuation applies.*
· *Nature and extent of work*	*Any limitations on inspection, enquiry, or analysis must be identified.*
· *Nature and sources of information*	*Significant information relied upon and verification controls.*
· *Significant assumptions and/or special assumptions*	*Any agreed special assumptions known prior to the valuation.*
· Specialist	***The use and role of a specialist. (NEW)***
· ESG factors	***Any requirements relating to ESG considerations. (NEW)***
· *Type of report being prepared*	*The format of the report must be described.*
· *Restriction of use, distribution and publication*	*Restrictions must be clearly communicated.*
· *IVS compliance*	*That the valuation will be prepared in compliance with IVS* ***and that the valuer will assess the appropriateness of all significant inputs.***

If, during the course of a valuation engagement, it becomes clear that the scope of work will not result in an IVS-compliant value, the valuation will not comply with IVS and this must be communicated to the client in writing.

IVS 102 – BASES OF VALUE

A basis of value describes the fundamental premises on which the reported values will be based. IVS permit a valuer to select any appropriate value definition, which could be specific to the purpose of the valuation, the country, local requirements, statues or regulations, etc. The selected basis (or bases) of value must be appropriate to the terms and purpose of the valuation assignment, as a basis of value may influence or dictate a valuer's selection of methods, inputs and assumptions, and the ultimate opinion of value.

A valuer is required to select the appropriate basis (or bases) of value and follow all applicable requirements associated with that basis of value, whether those requirements are included as part of IVS (for IVS-defined bases of value) or not (for non-IVS defined bases of value). The source of the definition of any basis of value used must be cited or the basis explained.

IVS-defined bases of value:

- **Market Value**
- ***Market Rent***
- **Equitable Value**
- **Investment Value/Worth**
- **Synergistic Value**
- **Liquidation Value**

"**Market Value** is the estimated amount for which an asset or liability should exchange on the valuation date between a willing buyer and a willing seller in an arm's length transaction, after proper marketing and where the parties had each acted knowledgeably, prudently and without compulsion." (definition from IVS 102, Appendix A10.1)

"**Market Rent** is the estimated amount for which an interest in real property should be leased on the valuation date between a willing lessor and a willing lessee on appropriate lease terms in an arm's length transaction, after proper marketing and where the parties had each acted knowledgeably, prudently and without compulsion." *(definition from IVS 102, Appendix A20.1)*

"**Equitable Value** is the estimated price for the transfer of an asset or liability between identified knowledgeable and willing parties that reflects the respective interests of those parties." *(definition from IVS 102, Appendix A30.1)*

"**Investment Value** is the value of an asset to a particular owner or prospective owner for individual investment or operational objectives. *This may also be known as "worth." (defined in IVS 102, Appendix A40.1)*

"**Synergistic Value** is the result of a combination of two or more assets or interests where the combined value is more than the sum of the separate values. If the synergies are only available to one specific buyer, then Synergistic Value will differ from Market Value, as the Synergistic Value will reflect particular attributes of an asset that are only of value to a specific purchaser. *The added value above the aggregate of the respective interests is often referred to as marriage value. (defined in IVS 102, Appendix A50.1)*

"**Liquidation Value** is the amount that would be realised when an asset or group of assets are sold on *from a liquidation sale, with the seller being compelled to sell as of a specific date. Liquidation value can be determined under two different premises of value: (a) an orderly transaction with a typical marketing period, or (b) a forced transaction with a shortened marketing period. (defined in IVS 102, Appendix A60.1)*

Forced Sale: *A forced sale is a description of the situation under which the exchange takes place, not a distinct premise of value. It is often used where a seller is under compulsion to sell, preventing a proper marketing period. The price obtainable in a forced sale reflects the seller's particular circumstances, rather than those of a hypothetical willing seller.*

IVS makes reference to four bases of value defined by other organisations.

- Fair Value **(IFRS):** IFRS 13 defines fair value as the price that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants at the measurement date. *(IVS 102, Appendix A70)Fair Value* ***(Legal/Statutory)*** *– Defined by courts in prior cases within individual jurisdictions. (IVS 102, Appendix A80)*

Valuers must choose the relevant basis or bases of value according to the terms and purpose of the valuation assignment. The choice of the basis of value should consider instructions and input received from the client and/or its representatives, however the valuer should not use a basis of value that is inappropriate for the intended purpose of the valuation. For bases of value defined by organisations other than the IVSC, it is the valuer's responsibility to understand and follow the regulation, case law and/or other interpretative guidance related to those bases of value as of the valuation date.

Premise of Value

A premise of value or assumed use describes the circumstances of how an asset and/or liability is used. Different bases of value may require a particular premise of value or allow consideration of multiple premises of value. IVS 102 describes and defines some common premises of value, including:

- Highest and best use
- Current use/existing use
- Orderly liquidation
- Forced sale

These premises are commonly, but inconsistently used across markets. IVS provides high level definitions of these terms to improve understanding and increase market consistency

Fair Value and Market Value Compared

Market Value (IVS)	Fair Value (IFRS)
• An estimated amount • It is time-specific and valid on the valuation date • The buyer is motivated to buy but, not overeager to buy at any price. • The seller is motivated to sell, but not forced to sell. • Arm's length transaction • The asset has been exposed to proper marketing • Parties to the transaction are reasonably informed about the nature, characteristics and uses of assets • The market for an asset could be an international market or a local market. • Does not reflect attributes of an asset that are valuable to a specific owner or purchaser • Disregard any synergies	• An estimated amount • Orderly transaction • Not a forced sale • Market based view • Not an entity specific measurement • Active Market • Highest and best use • Principal Market • Advantageous Market • Fair value hierarchy • Current price

Bangladesh Context

In Bangladesh "base of value" varies widely:

i. Bangladesh Bank:

FEID Circular No.- 1 (May 06, 2018) -Transfer of shares and repatriation of sale proceeds of shares in favor of nonresident in private/public limited companies not listed with stock exchanges:

"In accordance with the Guidelines for Foreign Exchange Transactions- 2018 (GFET) Bangladesh Bank would accept fair value of the shares as on the date of sale based on appropriate combination of three valuation approaches (viz. net asset value approach, market value approach and discounted cash flow approach) depending on the nature of the company."

As per the Circular, Fair Value shall be determined by Merchant Banker licensed by BSEC or a Chartered Accountant, for sales/transfer value of shares exceeding BDT 1.00 (One) million. Fair value has not been defined by Bangladesh Bank; hence, can be assumed "fair value" definition used in IFRS.

ii. Bangladesh Securities and Exchange Commission

BSEC Notification No. SEC/CMRRCD/2009-193/150/Admin…. (Aug 18, 2013), "The definition and source of the basis of value should be cited". It says basis of value should not contradict with applicable accounting standards.

This guideline is for revaluation of assets and "fair value" definition as per IFRS will be applicable.

iii. Income Tax Act 2023

- Sec 2(30) "Fair market value" means, in relation to capital asset-
  - (a) the price which such asset would ordinarily fetch on sale in the open market on the relevant day, and, where such price is not ascertainable, the price which the Deputy Commissioner of Taxes may, with the approval in writing of the Inspecting Joint Commissioner, determine;
  - (b) the residual value received from the lessee in case of an asset leased by a financial institution having license from the Bangladesh Bank on termination of lease agreement on maturity or otherwise subject to the condition that such residual value plus amount realized during the currency of the lease agreement towards the cost of the asset is not less than the cost of acquisition to the lessor financial institution.

  (Note that Fair market value" yet to be define under the Income Tax Act 2023)
- According to *Sec 107A (1)* arm's length price means a price in a transaction, the conditions (e.g. price, margin or profit split) of which do not differ from the conditions that would have prevailed in a comparable uncontrolled transaction between independent entities carried out under comparable circumstances.

iv. Value Added Tax & Supplementary Duty Act- 2012

According to Sec 2 (58): "fair market price" means–

(a) the consideration for a supply arrived at on the basis of a normal relation between a buyer and a seller, who are not associated with each other;

(b) if it is not possible to arrive at a fair market price as prescribed in clause (a) above, it would then be the consideration of a similar supply made previously under similar circumstances;

(c) if it is not possible to arrive at a fair market price by the above means, it may be determined by the Board on the basis of an objective approximation of considerations arrived at in the course of normal business relations among buyers and sellers, who are not associated with one another;

Entity-Specific Factors

IVS 102 includes a discussion of various factors that are specific to a particular buyer or seller but may not be available to participants generally, and the impact on the valuation. *Such factors include additional value from portfolio creation, unique synergies, entity-specific legal rights or restrictions, unique tax benefits or burdens, and abilities to exploit an asset that are unique to that entity.*

Synergies

IVS 102 defines synergies as the benefits associated with combining assets and/or liabilities. *When synergies are present, the value of a group of assets is greater than the sum of the values of the individual assets on a stand-alone basis. Whether synergies should be considered depends on the basis of value. For most bases, only those synergies available to other participants generally will be considered.*

Assumptions and Special Assumptions

IVS 102 differentiates between "assumptions" and "special assumptions". Assumptions are assumed facts that are consistent with or could be consistent with those existing at the date of the valuation, and special assumptions are those where the assumed facts differ from those existing at the valuation date or are different from what a typical participant would make ("hypothetical"). Special assumptions are so named to highlight to a user that the valuation conclusion is contingent on a change in the current circumstances, and are commonly used within the valuation of real property. All significant assumptions and special assumptions must be reasonable under the circumstances, be supported by evidence, and be relevant having regard to the intended use for which the valuation is required.

Transaction Costs

Most bases of value represent the estimated price of an asset without adjustment for the seller's costs of sale or the buyer's costs of purchase and any taxes payable by either party as a direct result of the transaction.

Allocation of Value

Allocation of value is the separate apportionment of value of an asset on an individual or component basis. When apportioning value, the allocation method must be consistent with the overall valuation premise/basis and the valuer must:

- *Follow any applicable legal or regulatory requirements*
- *Set out a clear description of the intended use of the allocation*
- *Consider the facts and circumstances, such as the relevant characteristic(s) of the item(s) being apportioned*
- *Adopt appropriate methodology(ies) in the circumstances*

IVS 103 – VALUATION APPROACHES

The principal valuation approaches are listed as "market", "income" and "cost", and IVS 103 includes information on the detailed methods of application for each approach. *A valuer is required to select the most appropriate method under the particular circumstances. The goal in selecting valuation approaches and methods is to find the most appropriate method under the particular circumstances of the valuation. Valuers are not required to use more than one method for the valuation, but should consider the use of multiple approaches and methods when there are insufficient factual or observable inputs for a single method to produce a reliable conclusion. Where multiple approaches and methods are used to arrive at a single conclusion, the valuer should analyze and reconcile the differing values obtained, and should describe within the valuation report this analysis and reconciliation. The process of analyzing and reconciling differing values into a single conclusion should be done* ***without averaging***.

IVS 103 acknowledges that compliance with IVS may require a valuer to use a method not defined or mentioned in the IVS.

The following approaches and methods are laid out in detail in IVS 103 and its Appendices, including the circumstances when they should be used and key steps in their application:

Market Approach

Market approach provides an indication of value by comparing the asset with identical or comparable (that is similar) assets and/or liabilities for which price information is available.

The market approach should always take into account trading volume, trading frequency, range of observed prices, and proximity to the valuation date.

Market approach methods include:

- **Comparable Transactions Method**

  Within this method, market information is collected from realized transactions with similar assets, and is combined with comparable evidence in order to calculate transaction multiples i.e. Units of comparison in business valuation can refer to: EBIT or EBITDA multiples, earnings multiples, book value multiples, etc.

Note that:

- Market information i.e. transaction prices is generally not available on the valuation date.
- *The valuer may also consider prices of identical or similar assets that are listed or offered for sale (comparable listings method), but this should not be used as the sole indication of value.*
- According to IVS 103, Appendix A10, the Valuer should make the following considerations when selecting comparable transactions:
- a choice of several transactions is generally preferable to a single transaction,
- transactions with very similar assets provide better indication of value than transactions with assets that require substantial adjustments,
- transactions closer to the valuation date are more representative; this depends also on market volatility and industry development
- information on transactions should come from reliable sources and should be sufficient to understand the background of the transactions
- actual transactions provide better evidence than intended transactions.
- *For most bases of value, transactions should be arm's length between unrelated parties*

The valuer should analyze and make adjustments for any significant differences between the comparable transactions and the subject asset, including:

- *Material characteristics (age, size, specifications)*
- *Geographical location*
- *Profitability or profit-making capability*
- *Historical and expected growth*
- *Differences related to marketability and control characteristics*
- *ESG considerations*
- *Ownership characteristics*

- **Guideline publicly-traded comparable method**

  Within this method, market information is collected from publicly-traded comparables that are the same as or similar to the subject asset. When valuing businesses, this method combines quoted prices of selected comparable companies, listed on the stock exchanges, with comparable evidence, in order to calculate guideline publicly-traded multiples. The units of comparison are the same as identified within the comparable transactions method.

  Under this method, market information, i.e. quoted prices, is available on the valuation date, and information on quoted companies is viewed as of higher reliability and availability, i.e. public filings have a certain degree of transparency, etc.

  According to IVS 103, appendix A10, the Valuer should:

  - *Choose publicly-traded comparables within the context of multiple comparables, similarity, and active trading*
  - *analyze and make adjustments for any material differences between the subject asset and the guideline publicly traded comparables, e.g. material characteristics, profitability, realized or expected growth, capital structure, any relevant restrictions on either the subject asset or on the comparable assets, Differences related to marketability and control characteristics, ESG considerations.*

  Other market approach considerations

  - "rule of thumb"
  - discounts for lack of marketability
  - control premiums
  - discount for lack of control
  - blockage discounts

Income Approach

Income approach provides an indication of value by converting future cash flows to a single current value. Under the income approach, the value of an asset is determined by reference to the value of income, cash flow or cost savings generated by the asset.

A fundamental basis for the income approach is that investors expect to receive a return on their investments and that such a return should reflect the perceived level of risk in the investment. Generally, investors can only expect to be compensated for systematic risk (also known as "market risk" or "undiversifiable risk").

Discounted Cash Flow Method

Under the DCF method, the forecasted cash flow is discounted back to the valuation date, resulting in a present value of the asset. In some circumstances for long-lived or indefinite-lived assets, DCF may include a terminal value.

When using the discounted cash flow method, according to IVS 2025 on, the Valuer should consider the following elements:

- Type of cash flows
- Cash flows to whole asset or a partial interest, e.g. cash flows to equity
- Pre-tax or post-tax cash flows
- Nominal versus real cash flows; real cash flows do not consider inflation
- Currency
- *Type of forecast.*

All of the above factors must be considered when selecting the appropriate type of cash flow for the nature of the asset or valuation. In addition, the discount rate and other inputs must be consistent with the type of cash flow chosen. (IVS 103, Appendix A20.05)

The valuer must consider all of the above factors and, additionally, use a discount rate appropriate for the defined cash flow, ensuring consistency between the cash flow type and the discount rate applied.

Explicit forecast period

Does not include mandatory requirements, however, offers guidance on the selection criteria for explicit forecast period (e.g. life of the subject asset, a reasonable period for which the data is available, the minimum period sufficient for an asset to achieve a stabilized level of growth and profits, for cyclical assets the forecast period should generally include an entire cycle, for finite-lived assets cash flows will typically be forecast over the full life). (IVS 103, Appendix A20.08-A20.11).

Cash flow forecasts

Cash flow forecasts are prepared on the basis of prospective financial information (PFI). "Regardless of the source of PFI, according to IVS 103, Appendix A20.13, the Valuer must perform an analysis to evaluate the PFI, the assumptions underlying the PFI and their appropriateness for the valuation purpose". *The projected cash flow should capture the amount and timing of all future cash inflows and outflows associated with the subject asset from the perspective appropriate to the basis of value. The valuer must ensure that seasonality and cyclicality have been appropriately considered. (IVS 103, Appendix A20.12-A20.19)*

Terminal value

If the asset is expected to continue beyond the explicit forecast, which is the case in most business valuations except if valuing the company under a liquidation assumption, the Valuer, according to IVS 103, Appendix A20.20, must estimate the value of the asset at the end of that period. *The terminal value is then discounted back to the valuation date, normally using the same discount rate as applied to the forecast cash flow.*

IVS 103 identifies 3 most often used terminal value estimation approaches, although indicating that the list provided isnon-exhaustive:

- Gordon growth model/ Constant growth model;
- Market approach / exit value; and
- Salvage value / disposal value. (IVS 103, Appendix A20.22-A20.28)

Discount rate

Does not include mandatory requirements, however, offers guidance on methods for estimating the discount rate, and indicates factors that, the Valuer should consider when developing the appropriate discount rate. The discount rate must be consistent with the type of cash flow. The valuer may use any reasonable method for developing an appropriate discount rate, including CAPM, WACC, observed rates/yields, or build-up methods. The valuer should consider corroborative analyses when assessing the appropriateness of a discount rate, including IRR, WARA, and value indications from other approaches. (IVS 103, Appendix A20.29-A20.40).

Cost Approach

Cost approach provides an indication of value using the economic principle that a buyer will pay no more for an asset than the cost to obtain an asset of equal utility, whether by purchase or by construction, unless undue time, inconvenience, risk or other factors are involved. *The approach provides an indication of value by calculating the current replacement or reproduction cost of an asset and making deductions for all relevant forms of obsolescence.*

Cost approach methods include:

- **Replacement cost method**

  Indicates value by calculating the cost of a similar asset offering equivalent utility. Replacement costs adjusted for different types of obsolescence.
- **Reproduction cost method**

  Indicates value by calculating the cost to recreating a replica of an asset. Example: Internally developed software that provides similar functions to the subject asset is, however, of current design and was developed using current cost-effective techniques and materials.
- **Summation method/underlying asset method**

  This method calculates the value of an asset by adding the separate values of its component parts. It is typically used for investment companies or other types of assets or entities for which value is primarily a factor of the values of their holdings.

Cost approach methods are primarily used for the valuation of real estate, plant & machinery and intangible assets. They are rarely applied for the valuation of businesses, except in above indicated situations. If applied, the summation method is used.

Cost Considerations

The cost approach should capture all costs that would be incurred by a typical participant, including:

- *Direct costs (materials, labour)*
- *Indirect costs (Transport, installation, professional fees, overheads, taxes, finance costs, profit margin)*

Depreciation / Obsolescence

In the context of the cost approach, "depreciation" refers to adjustments made to estimated cost to reflect the impact on value of any obsolescence affecting the subject asset. Depreciation adjustments are normally considered for:

- *Physical obsolescence: Loss of utility due to physical deterioration from age and usage (curable or incurable)Functional obsolescence: Loss of utility from inefficiencies in design, specification, or technology being outdated (excess capital cost or excess operating cost)*
- *External or economic obsolescence: Loss of utility caused by economic or locational factors external to the asset*

Is the Valuer required to always use more than one valuation method?

- *IVS 103, point 10.5, specifically states that the Valuer is not required to use more than one valuation method, especially in cases where the Valuer has a high confidence in the accuracy and the reliability of a single method.*
- *On the other hand, IVS 103 encourages the Valuer to consider the use of multiple approaches and methods when there are insufficient factual or observable inputs for a single method to produce a reliable conclusion.*

Whenever the Valuer uses multiple valuation approaches and/or methods, the Valuer should form the valuation conclusion based on analyzing and reconciling different valuation results into a single valuation conclusion, without averaging. (IVS 103, point 10.07specifically highlights the above issue).

What if different valuation approaches and methods result in widely divergent indications of value?

According to IVS 103, point 10.09, the Valuer should perform procedures to understand why the value indications differ, as it is generally not appropriate to simply weight two or more significantly divergent indications of value. In such cases, the valuer should reconsider the guidance in IVS 103, paragraph 10.04, to determine which valuation approach and/or method provides a better or more reliable indication of value.

Example: The Valuer appraised 100% of the equity of Company A using the market approach and the DCF, comparable transactions and guideline publicly-traded comparable methods and arrived at the following values for a 100% equity share i.e. majority and non-marketable: DCF = 1.5 million USD, comparable transactions method = 5 million EUR, guideline publicly-traded comparable method = 1.1 million EUR.

Prior to making the final valuation conclusion in the above example, the Valuer should analyze and review the comparable transaction calculation in order to understand why it indicates a value that is widely divergent from the DCF and guideline publicly-traded comparable method, e.g. size of the selected comparable transactions, arm's length transactions included, the effect of control premiums, synergies and other factors.

IVS 104 – DATA AND INPUTS

IVS 104 deals with the selection and use of data to be used as inputs in the valuation. The aim of the valuation is to maximize the use of relevant and observable data to the degree that it is possible. Data and inputs are used in developing values for all types of assets and liabilities.

IVS 104 applies to all valuation assignments and requires the valuer to be responsible for assessing and selecting the data, assumptions and adjustments to be used as inputs based upon professional judgement and professional scepticism.

- *Data and inputs should be based on factual information (such as measurements or published prices), but often include reasoning and analysis in order to arrive at an input to be used in the valuation.*
- *The valuation should maximize the use of observable data, which is defined as information that is readily available to market participants about actual events or transactions that are used in determining the value for the asset or liability.*
- *The valuer is responsible for assessing and selecting the data, assumptions and adjustments to be used as inputs in the valuation based upon professional judgement and professional scepticism.*

Use of a Specialist or Service Organisation

If the valuer does not possess all of the necessary data to perform all aspects of the valuation, it is acceptable for the valuer to engage a specialist or service organisation. Prior to using a specialist or service organisation, the valuer must ensure their capabilities meet the requirements of the intended use and must document their capabilities.

Characteristics of Relevant Data

The valuer must determine the data that is relevant, which for the purposes of IVS 104 means "fitness for use" in terms of the asset and/or liability being valued, the scope of work, the valuation method and the valuation model.

The valuer must apply professional judgement to balance the following characteristics of relevant data in order to choose the inputs used in the valuation:

- ***Accurate**– Data are free from error and bias and reflect the characteristics that they are designed to measure.*
- ***Complete**– The set of data is sufficient to address attributes of the assets or liabilities.*
- ***Timely**– Data reflect the market conditions as of the valuation date.*
- ***Transparent**– The source of the data can be traced from its origin.*

In certain cases, the data may not incorporate all of these characteristics. Therefore, the valuer must assess data and conclude, based on professional judgement, that the data is relevant to value the assets and/or liabilities in accordance with the scope of work and the valuation method.

Input Selection

Must be selected from relevant data in the context of the asset or liability being valued, the scope of work, the valuation method, and the valuation model. It must be sufficient for the valuation models being used to value the asset and/or liability based on the valuer using professional judgement. When valuing portfolios or groups of similar assets or liabilities, inputs should be selected appropriately across those portfolios or groups of assets. If significant inputs are inadequate or cannot be sufficiently justified, the valuation would not comply with IVS.

Data and Input Documentation

The source, selection and use of significant data and inputs must be explained, justified, and documented. Documentation must be sufficient to enable another valuer applying professional judgement to understand why specific data was determined to be relevant and inputs were selected and were considered reasonable. The form and location of documentation may vary based on the scope of work.

Environmental, Social and Governance (ESG) Considerations

The valuer should be aware of relevant legislation and frameworks in relation to the environmental, social and governance factors impacting a valuation. The impact of significant ESG factors should be considered in determining the value of a company, asset or liability. ESG factors may impact valuations both from a qualitative and quantitative perspective and may pose risks or opportunities that should be considered.

Examples of Environmental Factors:

- *Air and water pollution*
- *Biodiversity*
- *Climate change (current and future risks)*
- *Clean water and sanitation*
- *Carbon and other gas emissions*
- *Deforestation*
- *Natural disaster*
- *Resource scarcity or efficiency (e.g., energy, water and raw materials)*
- *Waste management*

Examples of Social Factors:

- *Community relations*
- *Conflict*
- *Customer satisfaction*
- *Data protection and privacy*
- *Development of human capital (health and education)*
- *Employee engagement*
- *Gender equality and racial equality*
- *Good health and well-being*
- *Human rights*
- *Working conditions*
- *Working environment*

Examples of Governance Factors:

- *Audit committee structure*
- *Board diversity and structure*
- *Bribery and corruption*
- *Corporate governance*
- *Donations*
- *ESG reporting standards and regulatory costs*
- *Executive remuneration*
- *Institutional strength*
- *Management succession planning*
- *Partnerships*
- *Political lobbying*
- *Rule of law*
- *Transparency*
- *Whistle-blower schemes*

ESG factors and the ESG regulatory environment should be considered in valuations to the extent that they are measurable and would be considered reasonable by the valuer applying professional judgement.

IVS 105 – VALUATION MODELS

IVS 105 addresses the selection and use of valuation models to be used in the valuation process. A valuation model is a tool used for the quantitative implementation of a valuation method in whole or in part, converting inputs into outputs used in the development of a value. This is distinct from a valuation method, which is a specific technique to develop a value.

IVS 105 applies to all valuation assignments and emphasizes that no model without the valuer applying professional judgement (for example, an automated valuation model or AVM) can produce an IVS-compliant valuation. The valuer must apply professional judgement and professional scepticism in the selection and use of valuation models and the application of inputs used in the valuation model.

- *Valuation models must be suitable for the intended use of the valuation and consistent with inputs.*
- *Valuation models can be developed internally or sourced externally from a specialist or service organisation.*
- *Valuation models used must be tested to ensure accuracy of the output is appropriate for the intended use, basis of value and the assets and/or liabilities being valued.*
- *In all cases, the valuer must apply professional judgement and professional scepticism in the selection and use of valuation models and the application of inputs.*

Use of a Specialist or Service Organisation

If the valuer does not possess all of the necessary valuation models to perform all aspects of the valuation, it is acceptable for the valuer to engage a specialist or service organisation to provide a valuation model. Prior to using a specialist or service organisation, the valuer must assess and document their capabilities.

Characteristics of Appropriate Valuation Models

The valuer must determine that the valuation model is appropriate, which for the purposes of IVS 105 means "fit for purpose" in terms of assets or liabilities being valued, the scope of work and the valuation method. The valuer must apply professional judgement to balance the following characteristics of a valuation model in order to choose the most appropriate valuation model:

- ***Accuracy** – The valuation model is free from error and functions in a manner consistent with the objectives of the valuation.*
- ***Completeness** – The valuation model addresses all the features of the asset and/or liability to determine value.*
- ***Timeliness** – The valuation model reflects the market conditions as of the valuation date.*

- ***Transparency*** *– All persons preparing and relying on the valuation model must understand how the valuation model works and its inherent limitations.*

In certain cases, the valuation model may not incorporate all of these characteristics. Therefore, the valuer must assess and conclude that the valuation model is appropriate to value the assets and/or liabilities in accordance with the scope of work and the valuation method.

Valuation Model Selection and Use

The valuation model should be selected in the context of the intended use, basis of value and the asset and/or liability being valued. Regardless of whether the valuation model is developed internally or externally sourced, the valuer must assess the valuation model in order to determine that it is fit for its intended use. The valuer must understand the way the valuation model operates. The valuation model should be tested for functionality and outputs must be analyzed for accuracy. Any significant limitations should be identified, along with any potentially significant adjustments.

Valuation models used over time should be maintained, monitored, assessed, and adjusted to ensure that they remain appropriate, accurate and complete. If significant limitations have been identified or adjustments required, then these must be explained, justified, and documented. If significant limitations or adjustments cannot be sufficiently justified, the valuation would not comply with IVS.

Valuation Model Documentation

A suitable valuation model should have documentation that includes the following information:

- *Support for the selection or creation of the valuation model.*
- *The purpose and intended use of the valuation model.*
- *The basis of value that the valuation model is designed to produce.*
- *The assumptions and limitations of the valuation model.*
- *The date of creation and any modifications of the valuation model.*
- *The inputs and data sources used by the valuation model.*
- *The outputs of the valuation model and how they are interpreted.*
- *Any testing or validation performed on the valuation model.*

Documentation should be sufficient to describe why the valuation model(s) were selected and considered by the valuer applying professional judgement.

IVS 106 – DOCUMENTATION AND REPORTING

IVS 106 establishes the requirements for documentation and reporting in valuations. Valuation reports and documentation are critical and defining features of IVS, which collectively assist in creating consistency, professionalism, transparency, comparability, and trust in valuation to serve the public interest.

IVS 106 applies to all valuation assignments and requires that an IVS-compliant valuation must have sufficient documentation and reporting to describe and provide transparency to the intended user on the valuation approach(es), valuation methods, inputs, valuation models, professional judgement, and resultant value(s).

- *The results of a valuation or valuation review must be documented and reported in writing and may include paper, electronic files, or other forms of recorded media.*
- *Documentation and reporting requirements apply regardless of whether the valuer is employed by the client or externally engaged by the client.*
- *Documentation must be maintained throughout the valuation and must describe the valuation and the basis of conclusions made.*
- *Reporting must be provided to the client in writing.*

Documentation

Documentation is the written record of the valuation or valuation review and may include communications with the client, working papers, or both, used to support the conclusions reached and compliance with IVS.

- *Documentation must be maintained to describe the valuation or valuation review and must be sufficient to describe the conclusion reached by the valuer.*
- *Documentation must be adequate to allow another valuer, applying professional judgement, to understand the scope of the valuation, the work performed, and the conclusions reached.*
- *In some cases, all documentation is included in the valuation report or valuation review report. In other cases, depending on the agreed scope of work, additional documentation must be maintained.*
- *Documentation should include but is not limited to communications with the client, alternative methods explored, additional data and inputs considered, risks and biases addressed, professional judgement used, and the valuation quality control procedures followed.*
- *In all cases, documentation should describe the valuation or valuation review and how the valuer managed valuation risk.*
- *The valuer must keep a copy of any report issued on the value and a record of the valuation work performed for a period in accordance with legal, regulatory, authoritative or contractual requirements relative to the intended use.*

Valuation Reports

Valuation reports must provide, in sufficient detail, a clear and well-structured description of the basis for the conclusion of value.

- *Valuation reports may reference other documents. These documents may include but are not limited to scope of work, internal policies, and procedures.*
- *Valuation reports should include all information necessary to provide the client with a clear description of the scope of work, the work performed, professional judgements made and the basis for conclusions reached.*
- *The format of the valuation reports may range from comprehensive narrative reports to abbreviated summary reports.*
- *Standing engagements that require frequent or repeated valuations may provide updates to an existing IVS-compliant report providing it is agreed upon in the scope of work.*

Valuation reports must convey the following, at a minimum:

- *The agreed scope of the work.*
- *The assets and/or liabilities being valued.*
- *The identity of the valuer.*
- *The client.*
- *The intended use.*
- *The intended users, if applicable.*
- *The valuation currency(ies) used.*
- *The valuation date(s).*
- *The basis/bases of value adopted.*
- *The valuation approach(es) adopted.*
- *The valuation method(s) or valuation model(s) applied.*
- *The sources and selection of significant data and inputs used.*
- *Significant environmental, social and governance factors used and considered.*
- *Any significant assumptions and/or special assumptions made.*
- *Any limitations on the use of the valuation or distribution of the report.*
- *The conclusion of value.*
- *The date of the valuation report.*
- *A statement that the valuation has been prepared in compliance with IVS.*

In all instances, the valuation report must be sufficient to describe the conclusion reached and be considered reasonable by the valuer applying professional judgement. If the valuer concludes that a limitation or restriction will impact compliance with IVS, the valuer must not state that the report is compliant with IVS.

Valuation Review Reports

A valuation review is not a valuation. A valuation review must state whether the review is a valuation process review or a value review or both:

- *A valuation process review addresses compliance with IVS.*
- *A value review addresses the reasonableness of a value.*

If a value is provided as part of the value review, then this is a valuation and must comply with the valuation reporting requirements of section 30 of this standard (IVS 106), including the minimum disclosures specified therein.

A valuation review must convey the following, at a minimum:

- *The agreed scope of the valuation review.*
- *The assets and/or liabilities reviewed.*
- *The identity of the valuation reviewer.*
- *The identity of the client.*
- *The intended use.*
- *The intended users, if applicable.*
- *Significant or special assumptions and/or limiting conditions pertaining to the valuation reviewed.*
- *The use of a specialist or service organisation if used, as part of the valuation review.*
- *Procedures undertaken and the documentation reviewed.*
- *The valuation reviewer's conclusions about the work under review, including supporting reasons.*
- *The subject of the review.*
- *The date of the valuation review report.*
- *The version of IVS that is applicable to the review.*

In all instances, the valuation review report must be sufficient to describe the conclusion reached and be considered reasonable by the valuer applying professional judgement.

ASSET STANDARDS

IVS include eight asset standards that apply depending on the type of asset being valued. The asset standards include requirements related to specific types of assets, including background information on the characteristics of each asset type that influence value and additional asset-specific requirements regarding common valuation approaches and methods used. The asset standards cover:

- **IVS 200 – Businesses and Business Interests**
- **IVS 210 – Intangible Assets**
- **IVS 220 – Non-Financial Liabilities**
- **IVS 230 – Inventory**
- **IVS 300 – Plant, Equipment and Infrastructure**
- **IVS 400 – Real Property Interests**
- **IVS 410 – Development Property**
- **IVS 500 – Financial Instruments**

THE ABOVE ASSET STANDARDS ARE PRECISELY EXPLAINED BELOW:

IVS 200 – BUSINESSES AND BUSINESS INTERESTS

IVS 200 contains additional requirements that apply to valuations of businesses and business interests. It is critical for the valuer to clearly define the business or business interest being valued. The type of value being provided must be appropriate to the purpose of the valuation and communicated as part of the scope of the engagement.

Business and business interests: The definition of what constitutes a business may differ depending on the purpose of a valuation. Generally, involves an organisation or integrated collection of assets engaged in commercial, industrial, service or investment activity. Generally, a business would include more than one asset (or a single asset in which the value is dependent on employing additional assets) working together to generate economic activity that differs from the outputs that would be generated by the individual assets on their own. *The absence of goodwill does not automatically imply that the asset or group of assets does not constitute a business.*

Businesses can take many legal forms, including but not limited to corporations, partnerships, joint ventures and sole proprietorships. Businesses can also include subsets or specific business activities of an entity, such as a division, a branch, or a segment.

The different levels at which value can be expressed are:

- **Enterprise value**

  Total value of equity plus the value of interest-bearing debt (debt and / or debt related liabilities) less cash and cash equivalents.

- **Total invested capital value**

  Value of total assets less current liabilities and cash

- **Operating value**

  Total value of the operations of the appraised business, excluding any value of non-operating (business non-related) assets or liabilities

- **Equity value**

  Value of business to all equity holders

Ownership Rights and Control Considerations (Controlling or non-controlling interest)

The Valuer would need to consider the "ownership rights" pertaining to the subject of the valuation.

The rights, privileges or conditions that attach to the ownership interest, whether held in proprietorship, corporate or partnership form, require consideration in the valuation. Ownership rights are usually defined within a jurisdiction by legal documents such as articles of association, memorandum of association, partnership agreements, shareholder agreements, and other corporate documents.

In some situations, the valuer may be required to distinguish between legal and beneficial ownership of a business interest. Corporate documents may contain restrictions on the transfer of an interest and/or other provisions relevant to value.

All rights and preferences associated with a subject business or business interest should be considered in a valuation, including but not limited to:

- *Where multiple classes of equity and/or hybrid securities exist, the valuation should consider the rights of each different class, including liquidation preferences, voting rights, redemption, conversion and participation provisions, and put and/or call rights.*
- *Where a controlling interest in a business may have a higher value than a non-controlling interest. Control premiums or discounts for lack of control may be appropriate depending on the valuation method(s) applied and/or the intended use of the valuation.*

Control Premium

A premium for control is required when the engagement is to value a controlling interest and the valuation yielded a non-controlling value. The International Glossary of Business Valuation Terms defines a control premium as "an amount or percentage by which the pro rata value of a controlling interest exceeds the pro rata value of a non-controlling interest in a business enterprise to reflect the power of control." When evaluating premiums paid in completed transactions, the valuer should consider whether the synergies and other factors that caused the acquirer to pay those premiums are applicable to the subject asset to a comparable degree.

Minority discounts

Minority discounts relate to the value of a particular shareholding where the rateable value (which reflects a control position) is discounted to reflect the lack of control over operations (discount for non-control). Discount for lack of marketability (DLOM) may also be applicable. The DLOM reflects the concept that when comparing otherwise identical assets, a readily marketable asset would have a higher value than an asset with a long marketing period or restrictions on the ability to sell the asset. Many bases of value allow the consideration of restrictions on marketability that are inherent in the subject asset but ***prohibit consideration of marketability restrictions that are specific to a particular owner****.*

DLOMs may be quantified using any reasonable method, but are typically calculated using option pricing models, studies that compare the value of publicly-traded shares and restricted shares in the same company, or studies that compare the value of shares in a company before and after an initial public offering.

The discount for lack of marketability generally relates to a minority shareholder position in a private company. However, care should be taken to avoid double-counting, as the en bloc value of a private company may already reflect certain marketability considerations.

In practice, discounts for non-control and lack of marketability are related but distinct concepts. The discount for non-control relates to the lack of control and reflects the relationship between the shareholders, while the discount for lack of marketability relates to the external market and the ability to sell the interest. ***In practice, these factors are sometimes combined and reflected in one discount, but the valuer should understand the conceptual distinction and ensure that any combined discount is supportable.***

VALUATION APPROACHES AND METHODS

IVS 200 provides a discussion of the three principal valuation approaches (market, income and cost) and how they may be applied to valuation of businesses and business interests. In addition to requirements already discussed in IVS 103, the Valuer should pay attention when assessing reasonableness of different factors.

Market Approach

The market approach is frequently applied in the valuation of businesses and business interests. The three most common sources of data used as inputs to value businesses and business interests using the market approach are:

- *Public markets in which ownership interests of similar businesses are traded.*
- *The acquisition market in which entire businesses or controlling interests in businesses are bought and sold.*
- *Prior transactions or offers for the ownership of the subject business.*

Factors that need to be considered may include:

- Similarity to the subject business in terms of qualitative and quantitative business characteristics, e.g., business description, EBIT/EBITDA margins, historical or projected sales growth, capital structure, quality of management, etc.
- Amount and verifiability of data on the similar business.
- Whether the price of the similar business represents a transaction consistent with the applicable basis of value (e.g., arm's length transaction; acquisition between group companies could probably not be treated as arm's length).
- Adjustments for control / lack of control and marketability.

When applying a market multiple, adjustments such as those for material characteristics, size, geographical location, profitability, growth, marketability, control characteristics, and ESG considerations may be appropriate.

Income Approach

- *The income approach is frequently applied in the valuation of businesses and business interests. Income and cash flow related to a business or business interest can be measured in a variety of ways and may be determined on a pre-tax or post-tax basis. The capitalisation or discount rate applied must be consistent with the type of income or cash flow used.*
- *The type of income or cash flow used must be consistent with the type of interest being valued:*
- *Enterprise value cash flows – Typically, cash flows before debt servicing costs, discounted with an appropriate discount rate applicable to enterprise-level cash flows, such as a weighted-average cost of capital (WACC).*
- *Enterprise Value Cash Flow = NOPAT + Depreciation and Amortization – CapEx – Change in Working Capital Requirement*
- *Equity value cash flows – Typically, cash flows after debt servicing costs, discounted with an appropriate discount rate applicable to equity-level cash flows, such as a cost of equity.*
- *Equity Value Cash Flow = Net Income + Depreciation and Amortization – CapEx – Change in Working Capital Requirement – Change in Debt Levels*
- *IVS encourages the valuer to review if adjustments, reflecting differences between realized historic cash flows and expected and recurring cash flows, are applicable:*
- *Adjusting revenues and expenses to levels that are reasonably representative of expected continuing operations.*
- *Adjusting non-arm's length transactions (such as contracts with customers or suppliers) to market rates, e.g., any compensations to employees that are above market norm and any special customer or supplier contracts.*
- *Adjusting the impact of non-recurring events, e.g., losses or gains on sale of company assets, remunerations for employees that were made redundant, and costs needed to start a new plant.*
- *Presenting financial data of subject and comparable companies on a consistent basis.*
- *Adjustments for control / lack of control and marketability.*
- *When using an income approach, it may also be necessary to adjust the valuation to reflect other matters that are not captured in either the cash flow forecasts or the discount rate adopted, such as adjustments for marketability or control. The valuer should ensure that adjustments to the valuation do not reflect factors that were already reflected in the cash flows or discount rate.*
- *While many businesses may be valued using a single cash flow scenario, the valuer may also apply multi-scenario or simulation models, particularly when there is significant uncertainty as to the amount and/or timing of future cash flows.*

Cost Approach

Cost approach is more often used for real estate, plant and equipment, or intangible asset valuations. It is rarely applied when valuing business interests, except in the following circumstances:

- *The business is an early stage or start-up business where profits and/or cash flow cannot be reliably determined and comparisons with other businesses under the market approach are impractical or unreliable.*
- *The business is an investment or holding business, in which case the summation method (underlying asset method) is applicable.*

  The business does not represent a going concern and/or the value of its assets and/or liabilities in a liquidation may exceed the business' value as a going concern.

Special Considerations for Businesses and Business Interests

IVS 200 also includes the following sections of topics (non-exhaustive) relevant to the valuation of businesses and business interests:

- **Ownership Rights** – All rights and preferences associated with a subject business or business interest such as restrictions on transfer of a business interest, multiple classes of stock, controlling Vs non-

controlling interests, etc., should be considered in a valuation. *Where multiple classes of equity and/or hybrid securities exist, the valuation should consider the rights of each different class.*

- **Business Information** – A valuer must assess the reasonableness of information received from management, representatives of management or other experts, and evaluate whether it is appropriate to rely on that information for the valuation purpose. *Although the value on a given valuation date reflects the anticipated benefits of future ownership, the history of a business may provide useful guidance to set expectations for the future.*
- **Economic and Industry Considerations** – Awareness of relevant economic developments and specific industry trends is essential for all valuations. *Matters including but not limited to political outlook, government policy, exchange rates, inflation, interest rates and market activity may affect assets and/or liabilities in different locations and/or sectors of the economy quite differently.*
- **Operating and Non-Operating Assets** – It is important to understand which assets and liabilities are required for use in the income-producing operations of the business and which ones are redundant or "excess" to the business at the valuation date. *Most valuation methods do not capture the value of assets and/or liabilities that are not required for the operation of the business. When separately considering non-operating assets and liabilities, the valuer should ensure that the income and expenses associated with non-operating assets and/or liabilities are excluded from the cash flow measurements and projections used in the valuation of the operating business.*
- **Capital Structure Considerations** – As businesses are often financed through a combination of debt and equity, the valuer should generally consider the allocation of the enterprise value between debt and any types of equity. *For simple capital structures that include only common stock and simple debt structures, it may be possible to estimate the value of all common stock by directly estimating the value of debt, subtracting that value from enterprise value, then allocating the residual equity value pro rata to all common stock.*

For complex capital structures that include forms of equity other than just common stock, the valuer may use any reasonable method to determine the value of equity or a particular class of equity. Three methods that the valuer may utilise in such instances are:

- ***Current Value Method (CVM)** – Allocates the enterprise value to various debt and equity securities assuming an immediate sale of the enterprise. The CVM should only be used when a liquidity event is imminent, or when the enterprise is at such an early stage that no significant common equity value above liquidation preference has been created.*
- ***Option Pricing Method (OPM)** – Values different share classes by treating each share class as an option on the cash flows from the enterprise. The OPM is often applied to capital structures in which the payout to different share classes changes at different levels of total equity value. The OPM most frequently relies on the Black-Scholes option pricing model.*
- ***Probability-Weighted Expected Return Method (PWERM)** – Estimates the value of various equity securities based upon an analysis of future values for the business, assuming various future outcomes. Share value is based upon the probability-weighted present value of expected future investment returns, considering each of the possible future outcomes.*

The valuer should consider recent transactions in the subject equity or a specific class of equity, and ensure the assumptions used are updated as necessary to reflect changes in the investment structure and changes in market conditions.

IVS 210 – INTANGIBLE ASSETS

IVS 210 contains requirements that apply to valuations of intangible assets. Intangible asset is defined as "a non-monetary asset that manifests itself by its economic properties. It does not have physical substance but grants rights and / or economic benefits to its owner."

IVS 210 lists the following categories of intangible assets:

- **Marketing-related** e.g. trademarks, trade names, internet domains etc.
- **Customer-related** e.g. customer lists, customer contracts, order backlog etc.

- **Artistic-related** e.g. books, films, plays, music etc.
- **Contract-related** e.g. licensing, royalty agreements, natural resource rights etc.
- **Technology-based** e.g. patented & unpatented technology, database, internally developed software etc.

Certain intangible assets, such as brands, may represent a combination of several categories listed above.

Goodwill is also addressed and generally defined, although the definition of goodwill differs depending on the purpose of a valuation. The value of goodwill can be different when calculated for different purposes, and for some purposes, goodwill may need to be further divided into transferable goodwill and non-transferable/ personal goodwill. *Since the amount of goodwill depends on which other tangible and intangible assets are recognised, its value can be different when calculated for different intended uses.*

The intangible assets that may need to be valued and how those intangible assets are defined may differ depending on the purpose of the valuation. The differences in how intangible assets are defined can lead to significant differences in value. IVS 210 recognizes that intangible assets may be valued for a variety of purposes, not just financial reporting purposes. It is the valuer's responsibility to understand the purpose of a valuation and whether intangible assets should be valued separately or grouped with other assets.

Circumstances requiring an intangible asset valuation include but are not limited to:

- *Financial reporting purposes, such as accounting for business combinations, asset acquisitions and sales, and impairment analysis.*
- *Tax reporting purposes, such as transfer pricing analyses, estate and gift tax planning and reporting, and ad valorem taxation analyses.*
- *Litigation in instances such as shareholder disputes, damage calculations and marital dissolutions (divorce).*
- *Other statutory or legal events such as compulsory purchases/eminent domain proceedings.*
- *General consulting, collateral lending, transactional support engagements and insolvency.*

Intangible assets can be valued for different purposes; however, in practice, financial reporting is one of the most common reasons, e.g. purchase price allocation, impairment testing, asset acquisitions, litigation and sales.

VALUATION APPROACHES AND METHODS

IVS 210 refers back to IVS 103 Valuation Approaches and Methods. Under each approach, IVS 210 provides guidance when each type of approach should be used, and examples of intangible assets for which the market, income or cost approaches are commonly used.

Market Approach

Under the market approach, the value of an intangible asset is determined by reference to market activity (for example, transactions involving identical or similar assets). *Transactions involving intangible assets frequently also include other assets, such as a business combination that includes intangible assets.*

The valuer should only apply the market approach to value intangible assets if both of the following criteria are met:

- *Information is available on arm's length transactions involving identical or similar intangible assets on or near the valuation date.*
- *Sufficient information is available to allow the valuer to adjust for all significant differences between the subject intangible asset and those involved in the transactions.*

The heterogeneous nature of intangible assets and the fact that they are seldom transacted separately from other assets limit the availability of market evidence. Where evidence is available, it usually comprises assets that are similar, but not identical, to the subject asset.

Examples of intangible assets for which the market approach is sometimes used include broadcast spectrum, internet domain names, and taxi licenses ("medallions").

Income Approach

Under the Income Approach, the following methods are discussed in some detail in the standard:

- **Excess earnings method**

  Estimates the value of an intangible asset as the present value of the cash flows attributable to the subject intangible asset after excluding the proportion of the cash flows that are attributable to other assets required to generate the cash flows (contributory assets). *The multi-period excess earnings method (MPEEM) is the most commonly used form as it offers the most flexibility and allows the valuer to explicitly forecast changes in inputs. Contributory asset charges (CACs) should be included for all current and future tangible assets, intangible assets and financial assets that contribute to the generation of the cash flow.*

- **Relief-from-royalty method**

  The value of an intangible asset is determined by reference to the value of the hypothetical royalty payments that would be saved through owning the asset, as compared with licensing the intangible asset from a third party. *The royalty rate may be based on market royalty rates for comparable transactions or on a split of profits that would hypothetically be paid in an arm's length transaction. Factors to consider include the competitive environment, importance of the asset to the owner, and the life cycle of the subject intangible.*

- **Premium profit method or with-and-without method**

  Indicates the value of an intangible asset by comparing two scenarios: one in which the business uses the subject intangible asset and one in which the business does not use the subject intangible asset. *This method is frequently used in the valuation of non-competition agreements. The comparison can be done by calculating the value of the business under each scenario or by calculating the difference in profits for each future period.*

- **Greenfield method**

  The value of the subject intangible is determined using cash flow projections that assume the only asset of the business at the valuation date is the subject intangible. All other tangible and intangible assets must be bought, built or rented. *This method is often used to estimate the value of "enabling" intangible assets such as franchise agreements and broadcast spectrum.*

- **Distributor method**

  Variation of the multi-period excess earnings method sometimes used to value customer-related intangible assets. *Information on profit margins earned by distributors is used to estimate the excess earnings attributable to customer-related intangible assets.*

Cost Approach

In most cases when valuing intangible assets under the cost approach, the replacement cost method will be used. The use of the reproduction cost method is limited because intangible assets do not have physical form. When applying the replacement cost method, the Valuers should consider:

- *The direct and indirect costs of replacing the utility of the asset, including labour, materials and overheads.*
- *Whether the subject intangible asset is subject to obsolescence (while intangible assets do not become physically obsolete, they can be subject to economic obsolescence).*
- *Whether it is appropriate to include a profit mark-up on the included costs.*
- *Opportunity costs that reflect costs associated with not having the subject intangible asset in place for some time during its creation.*

Special Considerations for Intangible Assets

IVS 210 also includes special considerations for intangible assets as follows (non-exhaustive list of topics relevant to valuing intangible assets):

Discount Rates/Rates of Return for Intangible Assets – Selecting discount rates for intangible assets can be challenging, as observable market evidence of discount rates for intangible assets is rare. The selection of a discount rate for an intangible asset generally requires significant professional judgement. When assessing the risks associated with an intangible asset, the valuer should consider factors including:

- *Intangible assets often have higher risk than tangible assets.*
- *If an intangible asset is highly specialised to its current use, it may have higher risk than assets with multiple potential uses.*
- *Single intangible assets may have more risk than groups of assets (or businesses).*
- *Intangible assets used in risky (non-routine) functions may have higher risk than those used in low-risk or routine activities.*
- *Intangible assets with longer lives are often considered to have higher risk, all else being equal.*
- *Intangible assets with more readily estimable cash flow streams, such as backlog, may have lower risk than similar intangible assets with less estimable cash flows, such as customer relationships.*

Benchmark rates to consider include: risk-free rates with similar maturities to the life of the subject intangible, cost of debt or borrowing rates with similar maturities, cost of equity rates or return for participants for the subject intangible assets, WACC of participants for subject intangible assets or of the company owning/using the subject intangible asset, internal rate of return (IRR) from recent business acquisitions, and weighted-average-return-on-assets (WARA) analysis.

- *Intangible Asset Economic Lives – An important consideration in the valuation of an intangible asset, particularly under the income approach, is the economic life of the asset. This may be a finite period limited by legal, technological, functional or economic factors; other assets may have an indefinite life. The economic life of an intangible asset is a different concept than the remaining useful life for accounting or tax purposes. Legal, technological, functional and economic factors must be considered individually and together in making an assessment of the economic life.*

  For customer-related intangible assets, attrition is a key factor in estimating both economic life and attributable cash flows. Attrition applied in the valuation of intangible assets is a quantification of expectations regarding future losses of customers. While it is a forward-looking estimate, attrition is often based on historical observations. There are several ways to measure and apply historical attrition, including constant rate of loss, variable rate of loss, and segregation of customers into different groups.
- *Tax Amortisation Benefit (TAB) – In many tax jurisdictions, intangible assets and in some cases, goodwill can be amortised for tax purposes. Depending on the intended use of a valuation and the valuation method used, it may be appropriate to include the value of the TAB in the value of the intangible asset and/or goodwill.*

  If the market or cost approach is used to value an intangible asset, the price paid to create or purchase the asset would already reflect the ability to amortise the asset. However, in the income approach, a TAB needs to be explicitly calculated and included, if appropriate.

  For some valuation intended uses, such as financial reporting, a TAB should be included when the income approach is used because a typical participant would be able to amortise an intangible asset acquired in such a hypothetical transaction. In calculating a TAB, the valuer may use either a discount rate appropriate for a business utilising the subject asset (such as WACC) or a discount rate appropriate for the subject asset itself.

IVS 220 – NON-FINANCIAL LIABILITIES

Non-Financial liabilities are defined as those liabilities requiring a non-cash performance obligation to provide goods or services. It includes: deferred revenue or contract liabilities, warranties, environmental liabilities, asset retirement obligations, certain contingent consideration obligations, loyalty programmes, power purchase agreements, certain litigation reserves and contingencies, and certain indemnifications and guarantees.

However, certain contingent consideration liabilities may require a non-cash performance obligation, such liabilities are not included in the scope of IVS 220 Non-Financial Liabilities.

VALUATION APPROACH AND METHODS:

The Market, Income and Cost Approach can all be applied to the valuation of non-financial liabilities.

Market Approach (Top-Down Method): Under the market approach, the value of a non-financial liability is determined by reference to market activity (for example, transactions involving identical or similar non-financial liabilities). While standalone transactions of non-financial liabilities are infrequent, valuers should consider relevant market-based indications of value.

Although such market-based indications may not provide sufficient information with which to apply the market approach, the use of market-based inputs should be maximised in the application of other approaches such as:

- Pricing from third parties
- Pricing for warranty policies issued by third parties
- The prescribed monetary conversion amount as published by participants for certain loyalty reward obligations
- The traded price for contingent value rights (CVRs)
- Observed rates of return for investment funds that invest in non-financial liabilities (eg, litigation finance).

Non-Financial liabilities seldom transact separately from other assets means that it is rarely possible to find market evidence of transactions

Where evidence of market prices is available, valuers should consider adjustments to these to reflect differences between the subject non-financial liability and those involved in the transactions. Such adjustments may only be determinable at a qualitative, rather than quantitative level.

A method to value non-financial liabilities under the Market Approach is often referred to as the Top-Down Method.

Top-Down Method: Under the Top-Down Method, valuing non-financial liabilities is based on the premise that reliable market-based indications of pricing are available for the performance obligation. A participant could theoretically price the liability by deducting costs already incurred toward the fulfilment obligation, plus a mark-up on those costs, from the market price of services

The key steps in the Top-Down Method are:

- *Determine the market price of the non-cash fulfilment.*
- *Determine the costs already incurred and assets utilised by the transferor.*
- *Determine a reasonable profit margin on the costs already incurred.*
- *Subtract costs incurred and profit from the market price.*

Income Approach Methods

The primary method to value non-financial liabilities under the Income Approach is often referred to as the Bottom-Up Method.

- **Bottom-Up Method:** Under this Method, the non-financial liability is measured as the costs (which may or may not include certain overhead items) required to fulfil the performance obligation, plus a reasonable mark-up on those costs, discounted to present value. It may be appropriate to include an assumed profit margin on certain costs which can be expressed as a target profit, either a lump sum or a percentage return on cost or value to Determine a reasonable mark-up. Determine the costs required to fulfil the performance obligation. When Determine timing of fulfilment and discount to present value. The discount rate should account for the time value of money and non-performance risk. *When fulfilment costs are derived through a percent of revenue, consider whether they already implicitly include the impact of discounting.*

Cost Approach

The cost approach has limited application for non-financial liabilities as participants typically expect a return on the fulfilment effort.

Special Considerations for Non-Financial Liabilities:

- Discount Rates for Non-Financial Liabilities
- Estimating Cash Flows and Risk Margins
- Restrictions on Transfer
- Taxes

Discount Rates for Non-Financial Liabilities

The discount rate should account for the time value of money and non-performance risk. Non-performance risk is typically a function of counterparty risk (i.e., credit risk of the entity obligated to fulfil the liability). Given the long-term nature of certain non-financial liabilities, the valuer should consider if inflation has been incorporated into the estimated cash flows and must ensure that the discount rate and cash flow estimates are prepared on a consistent basis.

Estimating Cash Flows and Risk Margins

Non-financial liability cash flow forecasts often involve the explicit modelling of multiple scenarios of possible future cash flow to derive a probability-weighted expected cash flow forecast. This method is often referred to as the Scenario-Based Method (SBM). The SBM also includes certain simulation techniques such as the Monte Carlo simulation. The SBM is commonly used when future payments are not contractually defined but rather vary depending upon future events. When the non-financial liability cash flows are a function of systematic risk factors, the valuer should consider the appropriateness of the SBM, and may need to utilise other methods such as option pricing models (OPMs).

In developing a risk margin, the valuer must document the method used and provide evidence for the computation. The cash flow risk margin should be the compensation that would be required for a party to be indifferent between fulfilling a liability that has a range of possible outcomes and one that will generate fixed cash outflows.

Restrictions on Transfer

Non-financial liabilities often have restrictions on the ability to transfer. Such restrictions can be either contractual in nature, or a function of an illiquid market for the subject non-financial liability. *When relying on market evidence, the valuer should consider an entity's ability to transfer such liabilities and whether adjustments to reflect the restrictions should be included.*

Taxes

The valuer should use pre-tax cash flows and a pre-tax discount rate for the valuation of non-financial liabilities. In certain circumstances, it may be appropriate to perform the analysis with after-tax cash flows and after-tax discount rates. In such instances, the valuer must explain the rationale for use of after-tax inputs.

IVS 230 – INVENTORY

Inventory broadly includes goods which will be used in future production processes (ie, raw materials, parts, supplies), goods used in the production process (ie, work-in-process), and goods awaiting sale (ie, finished goods).

This standard focuses on valuation of inventory of physical goods that are not real property. The valuation of real property is covered in IVS 400 Real Property Interests.

Valuations of inventory are performed for a variety of intended uses including financial reporting purposes (business combinations, asset acquisitions, impairment analysis), tax reporting purposes, litigation, general consulting, collateral lending, and insolvency.

VALUATION APPROACHES AND METHODS:

The three valuation approaches: market approach, cost approach and income approach can all be applied to the valuation of inventory

Market Approach:

The market approach, ie, reference to market activity involving identical or similar goods, has only narrow direct application for the valuation of inventory. Such applications typically include 1) inventory of commoditised products, or 2) inventory in which a market exists for the inventory at an interim stage in the production process. While the market approach is not directly applicable in most instances, valuers should consider market-based indications to determine the selling price as an input for other methods. *Other observable markets may provide insights on the returns attributable to the manufacturing and disposition of assets (e.g., distributor profit margins, contract manufacturer margins).*

Valuers should only apply the market approach to value inventory if both of the following criteria are met:

- Information is available on arm's length transactions involving identical or similar inventory on or near the valuation date, and
- Sufficient information is available to allow the valuer to adjust for all significant differences between the subject inventory and those involved in the transactions.

Income Approach

The valuation of inventory using the income approach requires the allocation of profit (value) contributed pre-valuation date versus the profit (value) contributed post-valuation date.

The primary method under the income approach is the Top-Down Method, which begins with the estimated selling price and deducts remaining costs and estimated profit. The key steps are:

- *Estimate the selling price (using direct observations or applying an appropriate gross profit margin).*
- *Estimate the costs to complete (for work-in-process only).*
- *Subtract the costs of disposal (selling and marketing expenses).*
- *Subtract the profit allowance on the completion effort (for work-in-process) and the disposal process.*
- *Consider any necessary holding costs (opportunity cost during the time required to sell).*

The Bottom-Up Method may also be used to corroborate the value derived from the Top-Down Method.

Cost Approach

The primary method to value inventory is the replacement cost method. Raw materials inventory is typically valued using the current replacement cost method.

Current Replacement Cost Method

The current replacement cost method (CRCM) may provide a good indication of market value if inventory is readily replaceable in a wholesale or retail business (eg, raw materials inventory).

The market value of raw materials and other inventory may be similar to the net book value as of the valuation date but certain adjustments should be considered:

- *The book value may need to be adjusted to FIFO basis.*
- *If raw material prices fluctuate and/or inventory turnover is slow, the book value may need to be adjusted for changes in market prices.*
- *The book value may be decreased to account for obsolete and defective goods.*
- *The book value may be decreased for shrinkage (difference between accounting records and actual inventory due to theft, damage, miscounting, etc.).*

The book value may be increased for any costs incurred in connection with raw material preparation (e.g., purchasing, storage and handling)

Special Considerations for Inventory

The following sections address a non-exhaustive list of topics relevant to the valuation of inventory:

- Identification of value-added processes and returns on intangible assets.
- Relationship to other acquired assets.
- Obsolete inventory.

- Unit of account-For purposes of inventory valuation, it is often appropriate to assume inventory is one homogenous set of assets.

IVS 300 – PLANT AND EQUIPMENT

IVS 300 now explicitly includes Infrastructure assets. Items of plant and equipment (which may sometimes be categorized as a type of personal property) are tangible assets that are usually held by an entity for use in the manufacturing/production or supply of goods or services, for rental by others or for administrative purposes and that are expected to be used over a period of time. *Infrastructure assets are typically part of a specialised system, network or group of complementary assets.*

Although Intangible assets fall outside the classification of plant and equipment assets but it may have an impact on the value of plant and equipment assets. Operating software, technical data, production records and patents are examples of intangible assets that can have an impact on the value of plant and equipment assets. *In such cases, the valuation process will involve consideration of the inclusion of intangible assets and their impact on the valuation*

A valuation of plant and equipment will normally require consideration of a range of factors including asset-related factors (technical specification, remaining useful life, condition, obsolescence), environmental or external factors (location, legislation, licences), and economic factors (profitability, demand for products).

VALUATION APPROACHES AND METHODS

The three principal valuation approaches may all be applied to the valuation of plant and equipment assets depending on the nature of the assets, the information available, and the facts and circumstances surrounding the valuation.

Market Approach

For classes of plant and equipment that are homogenous, eg, motor vehicles and certain types of office equipment or industrial machinery, the market approach is commonly used as there may be sufficient data of recent sales of similar assets. *When using the market approach, types of evidence include actual sales of identical assets, actual sales of similar assets, and asking prices. Actual sales must take preference over asking prices.*

Market evidence may be considered in a variety of ways including piecemeal (individual asset basis), production line, whole of plant/facility, or portfolio basis.

Adjustments may need to be considered for technical factors (size, capacity, specification), deterioration and obsolescence factors (condition, age, operating costs), market-related factors (location, currency, timing), and whether the asset is valued installed (in-situ) or removed (ex-situ).

Income Approach

The income approach to the valuation of plant and equipment can be used where specific cash flows can be identified for the asset or a group of complementary assets.

However, some of the cash flows may be attributable to intangible assets and difficult to separate from the cash flow contribution of the plant and equipment. *When plant and equipment is valued on an income approach, elements of value that may be attributable to intangible assets and other contributory assets should typically be excluded.*

Use of the income approach is not normally practical for many individual items of plant or equipment; however, it can be utilized in assessing the existence and quantum of economic obsolescence for an asset or asset group.

The income approach should be afforded significant weight when the asset or group has a high barrier to entry, significant time is involved to create an asset of equal utility, there are legal or regulatory hurdles, or a purchaser would pay a premium for immediate use.

Cost Approach

The cost approach is commonly adopted for plant and equipment, particularly in the case of individual assets that are specialized or special-use facilities. *The first step is to estimate the cost to a market participant of replacing the subject asset by reference to the lower of either reproduction or replacement cost.*

Cost-to-Capacity Method

Under the cost-to-capacity method, the replacement cost of an asset with an actual or required capacity can be determined by reference to the cost of a similar asset with a different capacity. *This method can be used as a primary method for determining replacement cost or as a check method. The relationship between cost and capacity is often not linear, so some form of exponential adjustment may be required.*

Trending Method

Trending is a method of estimating an asset's reproduction cost by applying an index (trend factor) to the asset's historical cost which reflects price inflation/deflation over time. Care should be taken when trending long-dated historical costs or using indices from different jurisdictions.

Special Considerations for Plant and Equipment

Generally, the value of an asset is independent of how it is financed. However, in some circumstances the way items of plant and equipment are financed and the stability of that financing may need to be considered in valuation.

Allocation of Value – Where a group of assets have been valued as part of a portfolio but allocated on an individual basis, the valuer must explicitly state this and provide rationale for the allocation methodology.

IVS 400 – REAL PROPERTY INTEREST

A real property interest is a right of ownership, control, use or occupation of land and buildings. A real property interest includes informal tenure rights for communal/community and or collective or tribal land and urban/rural informal settlements or transition economies, which can take the form of possession, occupation and rights to use.

Property interests are normally defined by state or the law of individual jurisdictions and are often regulated by national or local legislation. In some instances, legitimate individual, communal/community and/or collective rights over land and buildings are held in an informal, traditional, undocumented and unregistered manner. Before undertaking a valuation of a real property interest, a valuer must understand the relevant legal framework that affects the interest being valued.

Intangible assets may be associated with, and have a material impact on, the cash flows associated with real property assets. It is therefore essential to be clear in the scope of work precisely what the valuation is intended to include or exclude.

There are three main types of interest:

- the superior interest in any defined area of land
- a subordinate interest that normally gives the holder rights of exclusive possession and control of a defined area of land or buildings for a defined period, eg, under the terms of a lease contract
- a right to use land or buildings but without a right of exclusive possession or control, eg, a right to pass over land or to use it only for a specified activity

VALUATION APPROACHES AND METHODS

The following three valuation approaches can all be applicable for the valuation of a real property interest.

Market Approach

Property interests are generally heterogeneous (ie, with different characteristics). Notwithstanding these dissimilarities, the market approach is commonly applied for the valuation of real property interests.

In order to compare the subject of the valuation with the price of other real property interests, valuers should adopt generally accepted and appropriate units of comparison that are considered by

participants, dependent upon the type of asset being valued. *Units of comparison commonly used include price per square metre (or per square foot), price per hectare (or per acre), price per room, and price per unit of output.*

The valuer should consider differences between the subject property and comparable properties including: the type of interest, respective locations, quality of land, age and specification of improvements, permitted use or zoning, circumstances of the transaction, effective date of price evidence, and market conditions.

Income Approach

Various methods are used to indicate value under the general heading of the income approach, all of which share the common characteristic that the value is based upon an actual or estimated income that either is, or could be, generated by an owner of the interest.

For some real property interests, the income-generating ability of the property is closely tied to a particular use or business/trading activity (for example, hotels, golf courses, etc.). Where a building is suitable for only a particular type of trading activity, the income is often related to the actual or potential cash flows that would accrue to the owner of that building from the trading activity. *When the potential income represents cash flow from a business/trading activity and includes intangible assets, this is no longer solely a real property interest valuation and the valuer should also comply with IVS 200 and IVS 210 as appropriate.*

The discount rate in a discounted cash flow model will be based on the time cost of money and the risks and rewards of the income stream. If the objective is market value, the discount rate may be derived from observation of returns implicit in market transactions. If the objective is investment value, the rate may reflect the owner's required rate of return or weighted-average-cost-of-capital.

Cost Approach

It may be used as the primary approach when there is either no evidence of transaction prices for similar property or no identifiable actual or notional income stream that would accrue to the owner of the relevant interest. This approach is generally applied to the valuation of real property interests through the depreciated replacement cost method.

The replacement cost must reflect all incidental costs such as the value of the land, infrastructure, design fees, finance costs and developer profit that would be incurred by a participant in creating an equivalent asset. The cost must then be adjusted for physical, functional, technological and economic obsolescence.

Hierarchy of Interests

The different types of real property interests are not mutually exclusive. A superior interest may be subject to one or more subordinate interests. Lease interests granted directly by the owner of the absolute interest are "head lease" interests. The holder of a head lease interest can grant a sub-lease interest, which will always be shorter than, or coterminous with, the head lease out of which it is created.

When valuing a real property interest, it is necessary to identify the nature of the rights accruing to the holder of that interest and reflect any constraints or encumbrances imposed by the existence of other interests in the same property.

Rent

When valuing either a superior interest that is subject to a lease or an interest created by a lease, the valuer must consider the contract rent and, in cases where it is different, the market rent. The contract rent is the rent payable under the terms of an actual lease. It may be fixed for the duration of the lease or variable.

IVS 410 – DEVELOPMENT PROPERTY

This standard only includes modifications, additional requirements or specific examples of how the General Standards apply for valuations to which this standard applies. Valuations of development property must also follow IVS 400 Real Property Interests.

The residual value or land value of a development property can be very sensitive to changes in assumptions or projections concerning the income or revenue to be derived from the completed project or any of the development costs that will be incurred. If the valuation is required for an intended use where significant changes in value over the duration of a construction project may be of concern to the user, the valuer must highlight the potentially disproportionate effect of possible changes.

In the context of this standard, development properties are defined as interests where redevelopment is required to achieve the highest and best use, or where improvements are either being contemplated or are in progress at the valuation date and include:

- the construction of buildings,
- previously undeveloped land which is being provided with infrastructure,
- the redevelopment of previously developed land,
- the improvement or alteration of existing buildings or structures,
- land allocated for development in a statutory plan, and
- land allocated for a higher value uses or higher density in a statutory plan.

VALUATION APPROACHES AND METHODS

The valuation approach to be used will depend on the required basis of value as well as specific facts and circumstances. There are three main valuation approaches and one main valuation method: the market approach, the income approach, the cost approach, and the residual method (which is a hybrid of the market, income and cost approaches).

Market Approach

Some types of development property can be sufficiently homogenous and frequently exchanged in a market for there to be sufficient data from recent sales to use as a direct comparison where a valuation is required. The market approach may have limitations for larger or more complex development property, or smaller properties where the proposed improvements are heterogeneous. *For partially-completed development property, the application of the market approach is even more problematic as such properties are rarely transferred between participants in their partially-completed state.*

Income Approach

Establishing the residual value of a development property may involve the use of a cash flow model in some markets. The income approach may also be appropriate for establishing the value of a completed property as one of the inputs required under the residual method, which is explained more fully in the section on the residual method

Cost Approach

Establishing the development costs is a key component of the residual approach. The cost approach may also exclusively be used as a means of indicating the value of development property such as a proposed development of a building or other structure for which there is no active market on completion.

Residual Method

The residual method is normally a combination of market approach, income approach and cost approach. It indicates the residual amount after deducting all known or anticipated costs required to complete the development from the anticipated value of the project when completed after consideration of the risks associated with completion of the project.

The following basic elements should be considered in the application of the residual method:

- *Completed property value (Gross Development Value)*
- *Construction costs*
- *Consultants' fees*
- *Statutory fees*
- *Marketing costs*
- *Timetable and phasing of cash flows*

- *Finance costs*
- *Development profit (on both land and building)*
- *Contingency*
- *Discount rate*

The valuer must adopt one of two basic underlying assumptions for the value on completion: either based on values that are current on the valuation date (assuming the project had already been completed), or based on projected values at the anticipated date of completion.

Development profit should reflect the level of risk perceived by a prospective buyer on the valuation date, considering the stage the project has reached, whether a buyer or lessee has been secured, and the size and anticipated remaining duration of the project.

IVS 500 – FINANCIAL INSTRUMENT

A financial instrument is a contract that creates rights or obligations between specified parties to receive or pay cash or other financial consideration. Such instruments include but are not limited to, derivatives or other contingent instruments, hybrid instruments, fixed income, structured products and equity instruments. A financial instrument can also be created through the combination of other financial instruments in a portfolio to achieve a specific net financial outcome.

Valuation risk exists in the valuation of financial instruments. Throughout the valuation, procedures and controls must be put in place that enable valuation risk to be assessed and managed to help ensure that the value is appropriate for its intended use.

If the valuer does not possess the necessary technical skills, experience, data, models, or knowledge to perform all aspects of a valuation, the valuer should seek the assistance of a specialist or a service organisation providing this is agreed by the client and disclosed.

VALUATION APPROACHES AND METHODS

The various valuation methods used in financial markets are based on variations of the market approach, the income approach or the cost approach as described as under:

Market Approach

A price obtained from trading on a liquid exchange on, or very close to, the time or date of valuation is normally the best indication of the market value of a holding of the identical instrument. However, it may be necessary to make adjustments to the price information if the observed instrument is dissimilar to that being valued or if the information is not recent enough to be relevant.

Income Approach

The value of financial instruments may be determined using a discounted cash flow method. The terms of an instrument determine, or allow estimation of, the undiscounted cash flows. it is necessary to establish the appropriate discount rate.

Cost Approach

In applying the cost approach, valuers must follow the guidance contained in IVS 105 Valuation Approaches and Methods. Broadly, there are three cost approach methods:

Replacement cost method: a method that indicates value by calculating the cost of a similar asset offering equivalent utility,

Reproduction cost method: a method under the cost that indicates value by calculating the cost to recreating a replica of an asset, and

Summation method: a method that calculates the value of an asset by the addition of the separate values of its component parts.

DATA AND INPUTS

A broad range of data, assumptions, and adjustments are used in developing inputs used in valuations for financial instruments. Valuations should use observable data, such as published prices and yields, but may also require the use of assumptions and adjustments.

The valuer is responsible for assessing and selecting relevant data, assumptions, and adjustments to be used as inputs based upon professional judgement and professional scepticism. In circumstances where directly relevant data is not available and proxy data is used, the valuer must assess that the various instruments to be used as proxies are sufficiently comparable.

The characteristics of relevant data are:

- *Accurate – data are free from error and bias.*
- *Complete – the set of data is sufficient to address the attributes of the assets and/or liabilities.*
- *Timely – data reflect the market conditions as of the valuation date.*
- *Transparent – the source of the data can be traced from its origin.*

If significant inputs are inadequate or cannot be sufficiently justified, the valuation would not comply with IVS.

VALUATION MODELS

A valuation model is a quantitative implementation of a method in whole or in part that converts inputs into outputs used in the development of a value. A valuation model may be developed internally or sourced externally from a specialist or a service organisation.

The valuer must determine that the valuation model is appropriate, meaning "fit for use" in terms of assets and/or liabilities being valued, the scope of work, and the valuation method. The characteristics of appropriate valuation models are:

- *Accuracy – the valuation model is free from error and functions consistently with the objectives of the valuation.*
- *Completeness – the valuation model addresses all the features of the asset and/or liability.*
- *Timeliness – the valuation model reflects the market conditions as of the valuation date.*
- *Transparency – all persons preparing and relying on the valuation model must understand how it works and its inherent limitations.*

Valuation models must be tested prior to use, including appropriateness for intended use, suitability of inputs, mathematical accuracy, operational accuracy, and robustness. If testing reveals the model is not suitable, it must be remediated or rejected.

QUALITY CONTROL

Quality controls are procedures that ensure the valuation is performed consistently with IVS. Quality controls may be automated and/or manual and may include data reviews, valuation model validations, independent recalculation, back testing, and fact checking.

Quality controls should include a degree of review and challenge. Review and challenge should assess the process implemented and judgements made during the valuation, including review of work performed by specialists or service organisations.

For valuations with more complexity or involving multiple individuals or processes, a valuation control framework should be developed that includes:

- *Clear definition of roles and responsibilities of each party.*
- *Identification of responsible parties, including quality control and review and challenge.*
- *Valuation assessment, escalation, and remediation procedures.*
- *The types and extent of valuation risk associated with the valuation.*

Calibration must be performed during a valuation – a comparison of outputs from a valuation model with actual observed and/or expected outcomes. If the analysis produces evidence of inappropriate inputs or valuation model performance, action must be taken to address the issue.

CALIBRATION

Calibration must be performed during a valuation. Calibration is a comparison of outputs from a valuation model with actual observed and/or expected outcomes. If the analysis produces evidence of

inappropriate inputs or valuation model performance, action must be taken to address the nature of the issue and understand the causes of the variance.

Special Considerations for Financial Instruments

The following sections address a non-exhaustive list of topics relevant to the valuation of financial instruments.

- **Valuation Inputs** - Commonly used valuation input sources are broker quotations, consensus pricing services, the prices of comparable instruments from third parties and market data pricing services. *The valuer must be aware of market conventions such as quoted prices, spread or yield, ticks or basis points, and cash flow assumptions.*
- **Credit Risk** - Own credit and counterparty risk, credit risk of the instrument, credit risk of the issuer, Subordination, Leverage, Netting agreements, Default protection
- **Liquidity and Market Activity** -The liquidity of an asset is a measure of how easily and quickly it can be transferred in return for cash or a cash equivalent. Market activity is a measure of the volume of trading at any given time, and is a relative rather than an absolute measure.
- **Control Environment** - internal governance and control procedures that are in place with the objective of increasing the confidence of those who may rely on the valuation in the valuation process and conclusion. *Quality controls must be appropriately designed and executed, documented, and periodically assessed for recurring valuations.*

FUTURE CHANGES TO THESE STANDARDS

The IVSC Standards Board intends to continuously review the IVS and update or clarify the standards as needed to meet stakeholder and market needs. The Board has continuing projects that may result in additional standards being introduced or amendments being made to the standards in this publication at any time.

VALUATION AND ETHICAL PRINCIPLES

IVS based valuations require independent, unbiased and reliable valuations, with ethical conduct at the core of each valuation assignment. *Ethical conduct is fundamental to all valuation assignments performed under IVS. Valuers must adhere to the ethical principles set forth in the **IVSC Code of Ethical Principles for Valuers** (the most current version, updated from the original 2011 publication). These principles are generally binding for all professional valuers, depending on the regulatory development of the valuation profession in each country.*

*In addition to the ethical principles below, IVS 2025 explicitly introduces **Professional Scepticism** as a core valuer principle, requiring a questioning mind and critical assessment of valuation evidence at every stage of the valuation.*

The Code discusses the following five fundamental ethical principles, along with threats to compliance, safeguards, and steps a professional valuer should take:

1. **Integrity:** To be straightforward and honest in all professional and business relationships.
2. **Objectivity:** Not to allow conflict of interest, undue influence, or bias to override professional or business judgments.
3. **Professional Competence and Due Care:** To maintain the professional knowledge and skill required to ensure competent services based on current developments in practice, legislation, and techniques; to act diligently and in accordance with applicable technical and professional standards.
4. **Confidentiality:** To respect the confidentiality of information acquired as a result of professional and business relationships and not to disclose such information to third parties without proper and specific authority (unless there is a legal or professional right or duty to disclose), nor to use information for personal advantage. *This requires the valuer to maintain confidentiality of information received from clients and to ensure that staff and external advisors respect this duty.*

5. **Professional Behavior:** *To comply with relevant laws and regulations and to avoid any action that discredits the profession.*

IMPORTANT NOTE ABOUT THIS GUIDE DOCUMENT

This Guide Document is limited in scope and may not contain all matters required for consideration. It is intended to be a helpful tool. Practitioners are cautioned and reminded of their obligations to read and be familiar with the latest published version of IVS prior to accepting an engagement or providing valuation services in accordance with the IVS. IVS-related training is also available and should be considered by practitioners. IVS website can be accessed at https://www.ivsc.org/.